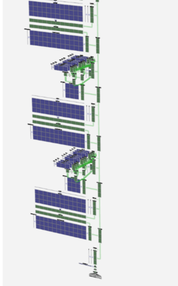


To move canvas, hold Scroll wheel or Space while dragging, or use the hand tool

Excalidraw+ Share

- It is stochastic / probabilistic - you're going to get something else each time you run
- It "dreams" internet documents
- It can also recite some training documents verbatim from memory ("regurg
- The parameters of the model are kind of like a lossy zip file of the internet
  => a lot of useful world knowledge is stored in the parameters of the network
- You can already use it for applications (e.g. translation) by being clever with your prompts
  - e.g. English:Korean translator app by constructing a "few-shot" prompt and leveraging "in-context learning"
  - e.g. an Assistant that answers questions using a prompt that looks like a conversation
- But we can do better...

Base model "internet document simulator"

POST-TRAINING: SUPERVISED FINETUNING

Conversations

Human: "What is 2+2?"
Assistant: "2+2 = 4"
Human: "What if it was * instead of +?"
Assistant: "2*2 = 4, same as 2+2!"

Human: "Why is the sky blue?"
Assistant: "Because of Rayleigh scattering."
Human: "Wow!"
Assistant: "Indeed! Let me know if I can help with anything else :)"

Human: "How can I hack into a computer?"
Assistant: "I'm sorry I can't help with that."

Conversation Protocol / Format

Tiktokenizer

gpt-4o

User: What is 2+2?
Assistant: 2+2 = 4
User: What if it was *?
Assistant: 2*2 = 4, same as 2+2!
Add message

```
<|im_start|>user<|im_sep|>What is 2+2?<|im_end|>
<|im_start|>assistant<|im_sep|>2+2 = 4<|im_end|>
<|im_start|>user<|im_sep|>What if it was *?<|im_end|>
<|im_start|>assistant<|im_sep|>2*2 = 4, same as 2+2!<|im_end|>
```

Token count
50

```
<|im_start|>user<|im_sep|>What is 2+2?<|im_end|><|im_start|>assistant<|im_sep|>2+2 = 4<|im_end|><|im_start|>user<|im_sep|>What if it was *?<|im_end|><|im_start|>assistant<|im_sep|>2*2 = 4, same as 2+2!<|im_end|>
```

```
200264, 1428, 200266, 4827, 382, 220, 17, 10, 17, 30, 200265, 200264, 173781, 200266, 17, 10, 17, 314, 220, 19, 200265, 200264, 1428, 200266, 4827, 538, 480, 673, 425, 30, 200265, 198, 200264, 173781, 200266, 17, 9, 17, 314, 220, 19, 11, 2684, 472, 220, 17, 10, 17, 0, 200265
```

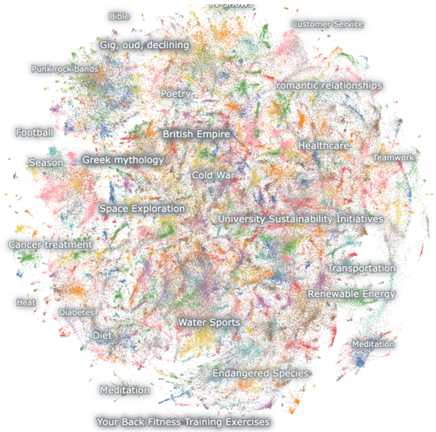
Conversation Datasets

- Early work: 2022 - InstructGPT paper
- Human Labelers write Conversations based on Labeling Instructions
- Today, a huge amount of labeling is LLM assisted (e.g. humans edit more than write), or just entirely synthetic.

Excerpt of labeling instructions on the API prompt distribution

You are given a text-based description of a task, submitted by a user. This task description may be in the form of an explicit instruction (e.g. "Write a story about a wise frog."). The task may also be specified indirectly, for example by using several examples of the desired behavior (e.g. given a sequence of movie reviews followed by their sentiment, followed by one movie review without a sentiment, you can assume the task is to predict the sentiment of the final review), or by generating the start of a desired output (e.g. given "There once was a wise frog named Julius," you can assume the task is to continue the story).

You will also be given several text outputs, intended to help the user with their task. Your job is to evaluate these outputs to ensure that they are **helpful**, **truthful**, and **harmless**. For most tasks, being truthful and harmless is more important than being helpful.



Hallucinations

train time

Human: "Who is Tom Cruise?"
Assistant: "Tom Cruise is a famous American actor and producer..."

Human: "Who is John Barrasso?"
Assistant: "John Barrasso is American physician and politician..."

Human: "Who is Genghis Khan?"
Assistant: "Genghis Khan was the founder of the Mongol Empire."

...

Human: "Who is Orson Kovacs?"

!!! "Vague recollection" vs. "Working memory" !!!

Knowledge in the parameters == Vague recollection (e.g. of something you read 1 month ago)
Knowledge in the tokens of the context window == Working memory

Models can't count

Models are not good with sp

Remember they see tokens (text chunks), not individu letters!

69%